Siliguri Institute Of Technology

Department : Master of Computer Application

STUDY OF URBANIZATION AND ITS EFFECT OVER SILIGURI USING SATELLITE IMAGERY

Presented by-

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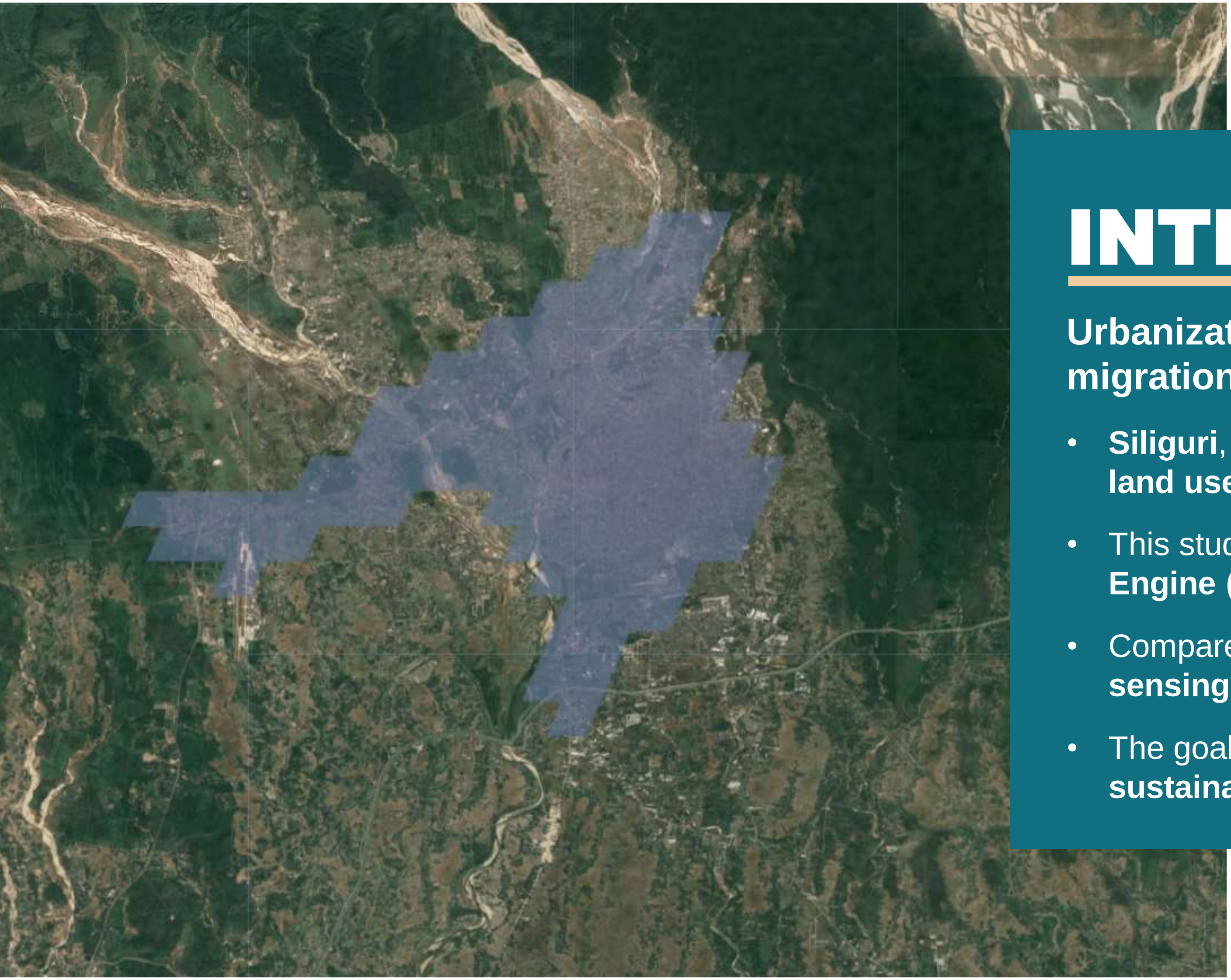


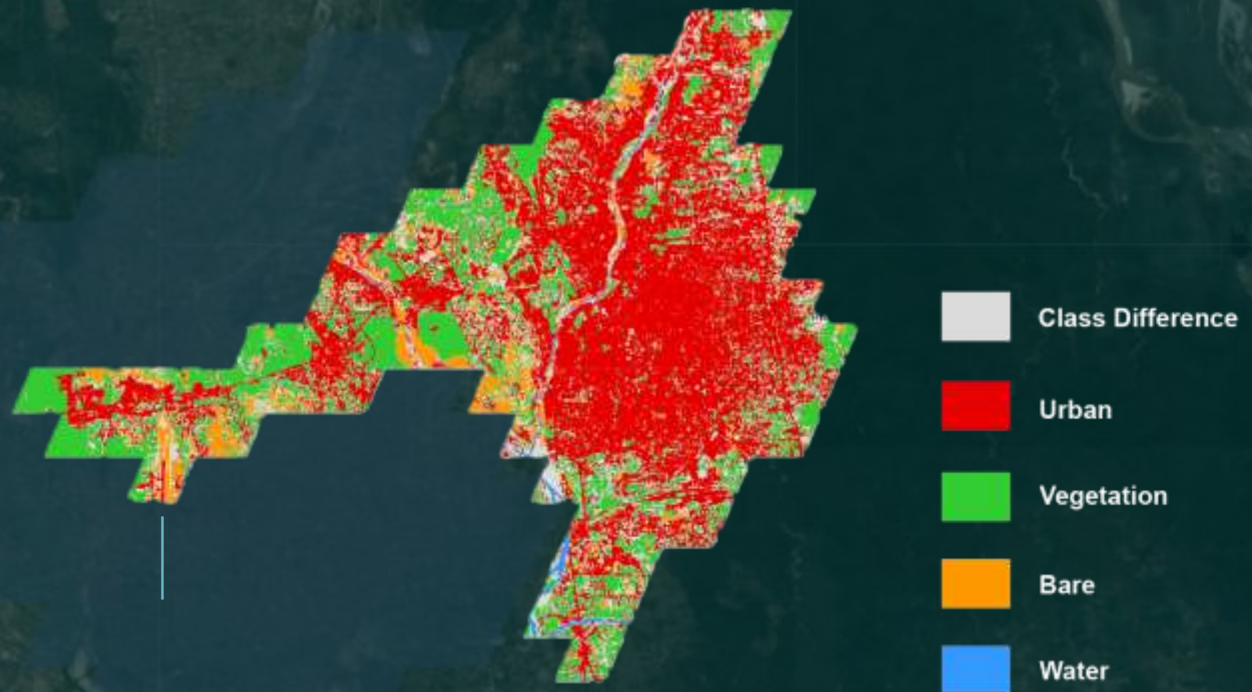
INTRODUCTION

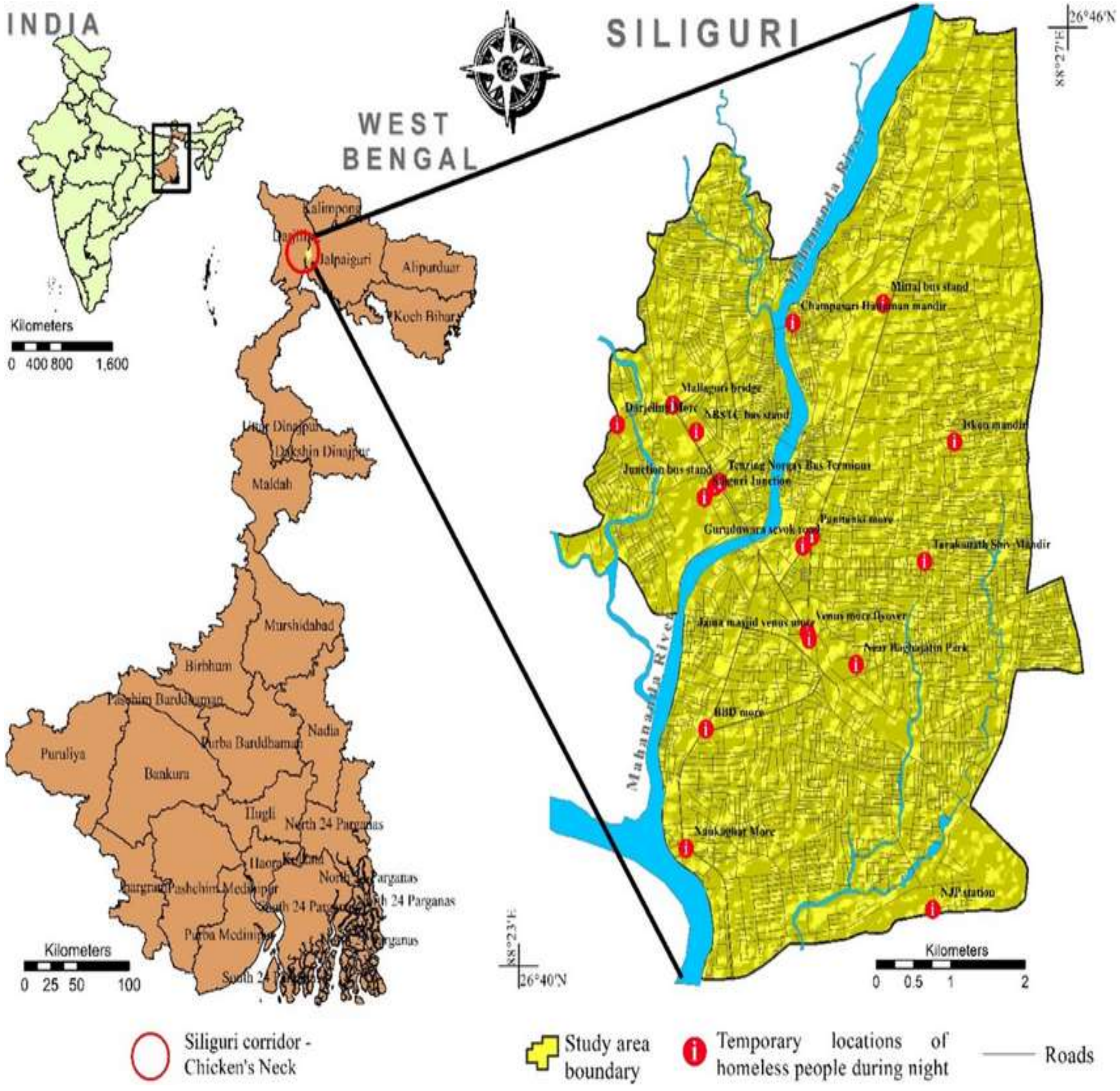
Urbanization is rapidly transforming cities worldwide, driven by **migration**, **economic growth**, and **infrastructure development**.

- **Siliguri**, the gateway to Northeast India, has seen major **land use changes** in recent years.
- This study uses **satellite imagery** and **Google Earth Engine (GEE)** to monitor urban growth in Siliguri.
- Compared to traditional ground surveys, this **remote sensing** method is **faster**, **scalable**, and **more accurate**.
- The goal is to support **efficient planning** and promote **sustainable development**.

OBJECTIVE

- Track **urban growth** in Siliguri using satellite data
- Monitor **land use changes** from 2020 to 2024
- Study the **loss of greenery and water bodies**
- Use **Google Earth Engine (GEE)** and **Sentinel-2 images**
- Create **maps** to show changes in city, vegetation, and water
- Help in **better planning** and **sustainable development**



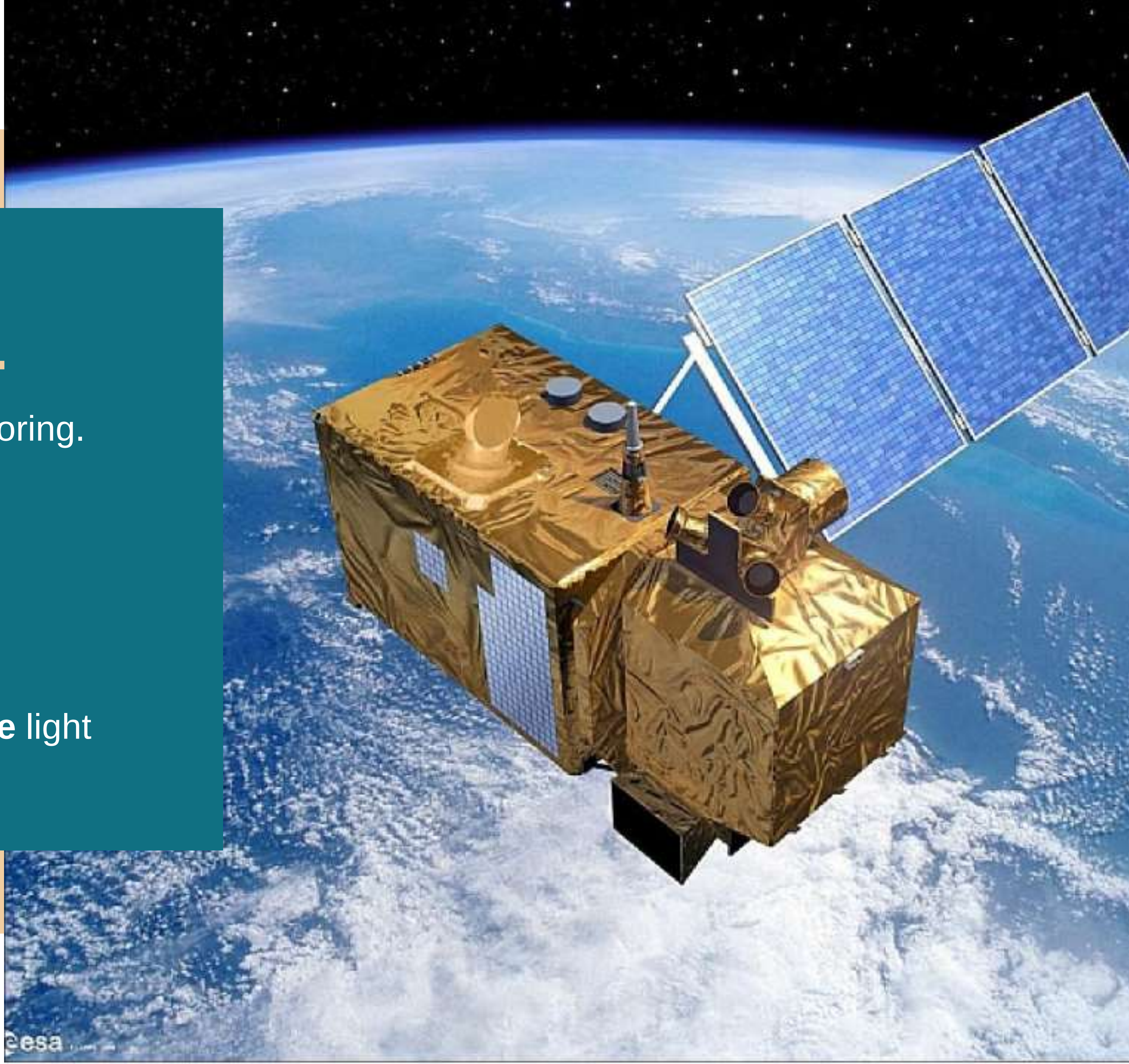


STUDY AREA – SILIGURI

- Siliguri is located in **northern West Bengal**
- Acts as a **commercial hub** and **transport gateway** to Northeast India
- **Rapid Urban Growth**
- Fast population rise
- Expansion of buildings, roads, and industries

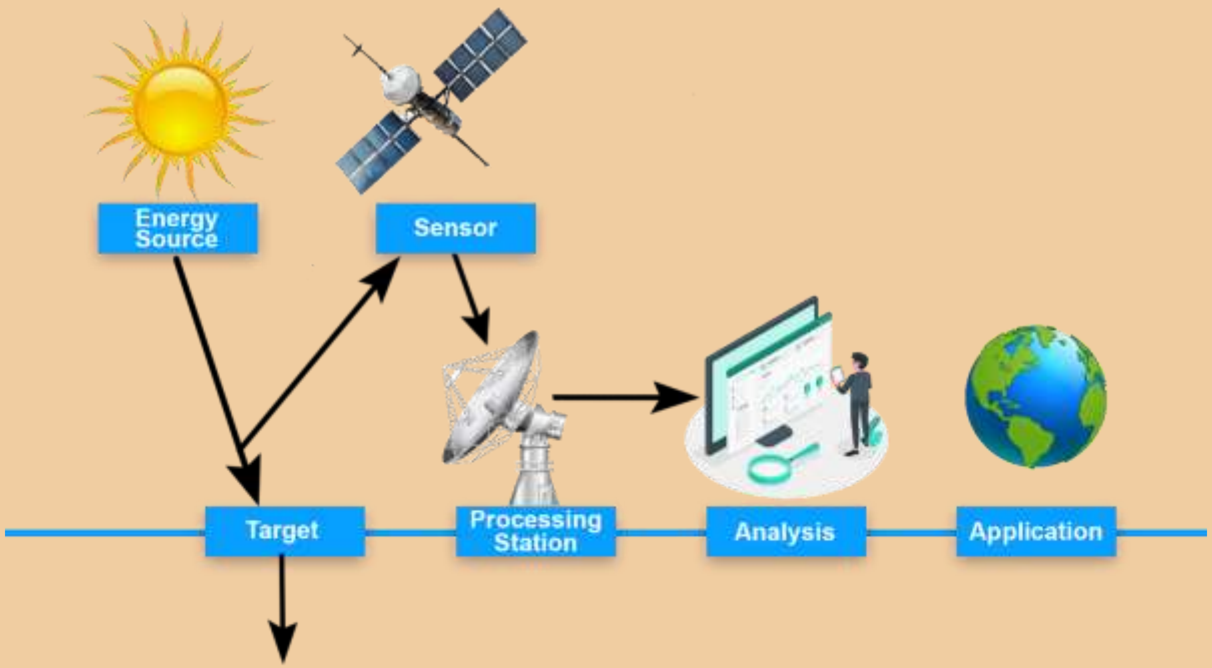
USING SATELLITE

- We used **Sentinel-2** satellite for monitoring.
- High-resolution **multispectral images**
- Free and regularly updated
- Ideal for:
  - -Urban development tracking
  - -Green cover monitoring
  - -Water resource analysis
- It captures **visible** (RGB) and **invisible** light (NIR, SWIR)



HOW SATELLITES CAPTURE EARTH'S IMAGES (REMOTE SENSING)

- **Energy Source**: The **Sun** sends light energy to Earth.
- **Target**: The **Earth's surface** (buildings, trees, water, etc.) reflects this energy.
- **Sensor**: The **satellite sensor** captures this reflected energy (visible + invisible light).
- **Processing Station**: The data is sent to a **ground station**, cleaned, and turned into usable images.
- **Analysis**: The images are analyzed on computers to detect changes (urban growth, vegetation loss, etc.).
- **Application**: The results help in **city planning, environment monitoring**, and decision-making.



TOOLS & DATA USED

Platform:

- **Google Earth Engine (GEE)**
  - Cloud-based platform for satellite data analysis
  - Used for land cover classification & change detection

Satellite Data:

- **Sentinel-2 Imagery (2020–2025)**
  - Multi-spectral images for monitoring urban, vegetation & water changes

Indices Used:

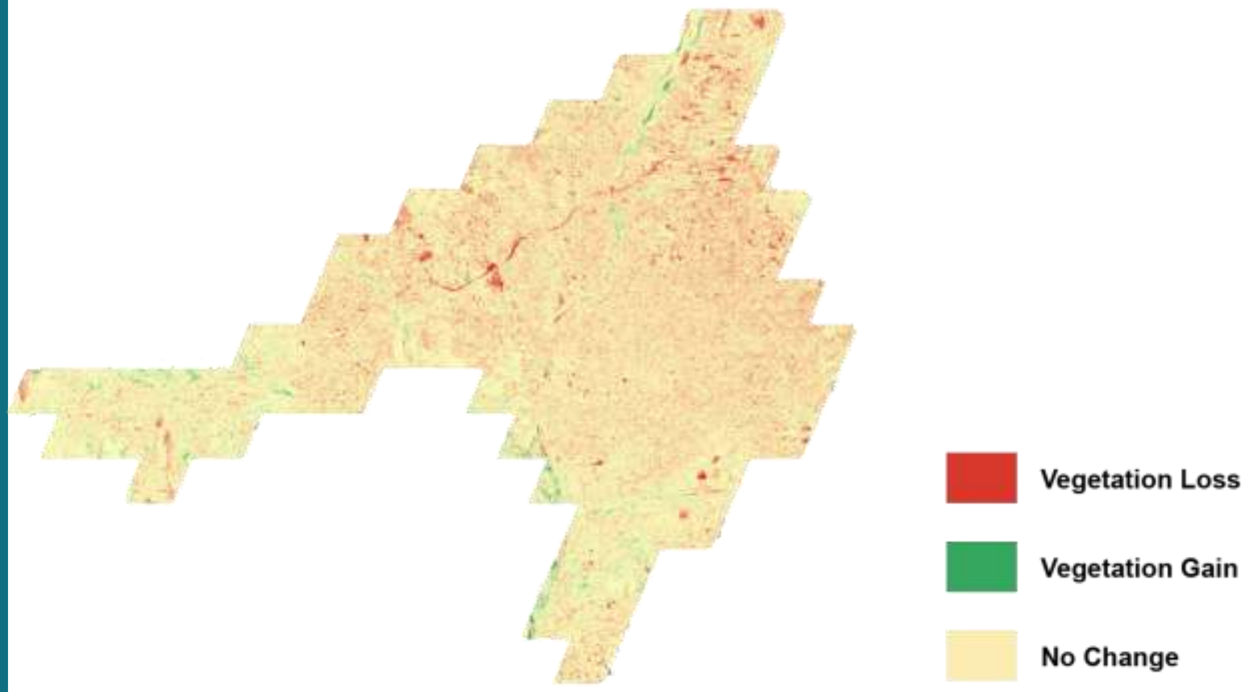
- **NDVI** – Detects vegetation health
- **NDBI** – Identifies built-up/urban areas
- **MNDWI** – Highlights water bodies

Software:

- **QGIS** – Used for map layout, visualization & spatial analysis

RESULTS & DISCUSSION

- **Urban areas increased** → More buildings, less open space
- **Vegetation decreased** → Green areas turned into concrete
- **Water bodies reduced** → Lakes and rivers shrinking
- **Environmental impact** → More heat, flooding, and pollution risks



FUTURE SCOPE

- **Use Landsat & ALOS**
  Add more satellite data for better accuracy and long-term tracking
- **Yearly Change Monitoring**
  Track built-up, vegetation, and water changes each year
- **Seasonal Analysis**
  Compare images across seasons for consistent results
- **Apply Deep Learning**
  Use smart models for accurate, automated land classification

THANK YOU